

Sourabh Vasant Gothe

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Professional Summary

Chief Machine Learning Engineer bridging the gap between rigorous theoretical innovation and productization. Specializing in NLP (Language Models, Knowledge Graphs), Computer Vision (Image/Video understanding) and on-device AI productization. Inventor and author with a portfolio of 10+ patents (8 A1-grade) and 10+ publications in top-tier venues (AAAI, WACV, ICASSP). Proficient in C++, Python, and algorithms, with a proven track record of architecting highly optimized, shipping production-ready AI systems. Active reviewer for NeurIPS, EMNLP, and WACV, passionate about driving deep-tech research into tangible product features.

Core Skills

- **Research & Innovation Vision:** Core problem identification, novel algorithm design under extreme hardware constraints, patent drafting (10+ portfolio), top-tier academic publishing (AAAI, WACV, ICASSP).
- **AI/ML Domains:** Agent Memory, RAG (Adaptive RAG, GraphRAG), Agentic Pipelines, Quantization/Pruning, LoRA, and On-Device Inference, User Personalization. Deep Learning, Computer Vision (Image/Video Understanding/3D Vision), Natural Language Processing (SLM/LLM/Knowledge Graphs), Generative AI (VAE/GAN/Diffusion, LRMs, CLIP/BLIP2),
- **Programming Languages:** C++, Python, Java/Kotlin.
- **Frameworks & Libraries:** PyTorch, LangGraph, Ollama, ONNX, TFLite, Hugging Face (peft).

Experience

Chief Engineer, Machine Learning | SAMSUNG R&D – Bangalore, India | 03/2024 - Present

- Leading a research and development initiative on agentic memory and Knowledge Graph-based personalization.
- Led research project focused on Lifelog Intelligence using Knowledge Graphs on mobile devices to enhance user personalization experience. Successfully published our research findings in AAAI Main Track-2026.
- Architected and deployed a training-free, on-device physics engine converting 2D images to 3D dynamics (ICASSP 2025), optimizing performance for strict mobile constraints.
- Engineered an end-to-end vision model (one-stage anchor-free object detection) for aesthetic widget placement, successfully commercialized and currently serving across Samsung Galaxy S25, S26, and other flagship devices.

Lead Engineer, Machine Learning | SAMSUNG R&D – Bangalore, India | 03/2021 - 01/2024

- Developed a highly optimized temporal segmentation deep learning architecture utilizing only 3M trainable parameters (22× fewer than SOTA) while achieving a 13.92% accuracy improvement (published in ICASSP 2023).
- Owned the end-to-end product lifecycle—from data curation and DNN modeling to on-device inference optimization—for a novel hybrid video-understanding algorithm (WACV 2024, ICASSP 2024).
- Successfully commercialized the temporal segmentation solution for production, directly integrating it into Samsung Galaxy S24 and S25 flagship devices.
- Designed a patented Emoji Drawing Detection system and developed an end-to-end native C++ neural inference engine for CNN-based models (CVIP 2022).

Senior Software Engineer | SAMSUNG R&D – Bangalore, India | 02/2019 - 03/2021

- Developed and deployed an on-device multilingual language model for the Indian market, enabling real-time language detection, code-switching, and macaronic language modeling.

- Engineered a lightweight Language Detection Engine (ICSC 2020) and a comprehensive on-device grammatical error correction system optimized for mobile constraints (ICSC 2021). Commercialized native C++ inference engines and NLP features into the Samsung Neural Keyboard, deployed globally across Galaxy flagship devices.

Software Engineer | SAMSUNG R&D – Bangalore, India | 07/2017 - 03/2019

- Trained and optimized N-gram and LSTM-based neural language models for multiple Indian and global languages, successfully commercialized for global market deployment.
- Implemented advanced data structures, including MARISA and CEDAR TRIE, to achieve highly efficient on-device user language modeling.

Research Intern | SAMSUNG R&D – Bangalore, India | 01/2017 - 07/2017

- Developed an HMM-based machine learning algorithm for human activity detection, specifically optimized for sensory data on low-resource devices.

Education

M.Tech, Data Science and Business Analytics | Indian Institute of Science (IISc), Bangalore | 07/2023

GPA: 9.3 | M.Tech (Online) Sponsored by Samsung Research

Bachelor of Engineering, Computer and Information Sciences | PES Institute of Technology, Bangalore | 06/2017

GPA: 9.13

Selected Publications

- [ContextGraph: Lifelog Intelligence Framework for Contextual Subgraph Evolution](#) | Anil Sharma, Gunturi Kiran, Jayesh Vachhani, **Sourabh Gothe**, Ayon Chattopadhyay, Yashwant Saini, Parameswaranath, Barath Raj. AAAI 2026.
- [PhysID: Physics-Based Interactive Dynamics From A Single-View Image](#) | **Sourabh Gothe**, Ayon Chattopadhyay, Gunturi Kiran, Pratik, Vibhav Agarwal, Jayesh Vachhani. IEEE ICASSP 2025 (**Bronze Award, Samsung Best Paper**).
- [What's in the Flow? Exploiting Temporal Motion Cues for Unsupervised Generic Event Boundary Detection](#) | **Sourabh Vasant Gothe**, Vibhav Agarwal, Sourav Ghosh, Jayesh Vachhani, Pranay Kashyap. IEEE/CVF WACV 2024.
- [SAM-GEED: Zero-Cost Approach for Generic Event Boundary Detection](#) | Pranay Kashyap, **Sourabh Gothe**, Jayesh Vachhani, Vibhav Agarwal, IEEE ICASSP 2024.
- [Self-Similarity Is All You Need For Fast And Light-Weight Generic Event Boundary Detection](#) | **Sourabh Gothe**, Jayesh Vachhani, Rishabh Khurana, Pranay Kashyap. IEEE ICASSP 2023.

Selected Patents

- [Method and electronic device for detecting AI generated content in a video](#) | A1 Grade
- [System and Method for Generating contextual transition based on guided denoising of flow](#) | A1 Grade
- [Method and electronic device for predicting plurality of multi-modal drawings](#) | A1 Grade
- [System and Method for Intelligent Temporal Segments based on Class Agnostic Video Networks](#) | A1 Grade

Portfolio includes 10+ patents (8 A1-grade) and 10+ publications in top-tier venues | Please check the portfolio

Awards & Recognitions

- Samsung Best Paper Awards 2024 | **Bronze Award**
- Samsung Annual Award 2023 | **Eminent Author of the Year**
- Samsung Excellence Award 2021, 2022, 2023 | **Star of the Quarter**
- Samsung Software Competency Level | Professional (Achieved with a thorough understanding of advanced DSA)